

● EASY

● ALGEBRA

● ANSWER KEY

Algebra

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01

Q1**5**

The correct answer is 5. Let p represent the number of packages of dinner rolls that should be bought for the party. It's given that dinner rolls are sold in packages of 12. Therefore, $12p$ represents the number of dinner rolls that should be bought for the party. It's also given that 50 dinner rolls are needed; therefore, $12p \geq 50$. Dividing both sides of this inequality by 12 yields $p \geq \frac{50}{12}$, or approximately $p \geq 4.17$. Since the number of packages of dinner rolls must be a whole number, the minimum number of packages that should be bought for the party is 5.

Q2**B****B.** 770

Choice B is correct. It's given that the equation $x + y = 1,440$ represents the number of minutes of daylight, x , and the number of minutes of non-daylight, y , on a particular day in Oak Park, Illinois. It's also given that this day has 670 minutes of daylight. Substituting 670 for x in the equation $x + y = 1,440$ yields $670 + y = 1,440$. Subtracting 670 from both sides of this equation yields $y = 770$. Therefore, this day has 770 minutes of non-daylight.

Choice A is incorrect. This is the number of minutes of daylight, not non-daylight, on this day.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the total number of minutes of daylight and non-daylight.

Q3**D**

D. 60

Choice D is correct. It's given that $5x = 20$. Multiplying both sides of this equation by 3 yields $15x = 60$. Therefore, the value of $15x$ is 60.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Q4**B**

B. 40

Choice B is correct. For the graph shown, the x -axis represents time, in seconds, and the y -axis represents the number of candy bars wrapped. The slope of a line in the xy -plane is the change in y for each 1-unit increase in x . It follows that the slope of the graph shown represents the estimated number of candy bars the machine wraps with a label per second. The slope, m , of a line in the xy -plane can be found using any two points, x_1, y_1 and x_2, y_2 , on the line and the slope formula $m = \frac{y_2 - y_1}{x_2 - x_1}$. The graph shown passes through the points 0, 0 and 2, 80. Substituting these points for x_1, y_1 and x_2, y_2 , respectively, in the slope formula yields $m = \frac{80 - 0}{2 - 0}$, which is equivalent to $m = \frac{80}{2}$, or $m = 40$. Therefore, the estimated number of candy bars the machine wraps with a label per second is 40.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q5**B**

B. $f + r = 214$

$$11f + 8.25r = 2,145$$

Choice B is correct. The movie theater sells f full-price tickets and r reduced-price tickets, so the total number of tickets sold is $f + r$. Since the movie theater sold a total of 214 full-price and reduced-price tickets for one movie showing, it follows that $f + r = 214$. The movie theater charges \$11 for each full-price ticket; thus, the sales for full-price tickets, in dollars, is given by $11f$. The movie theater charges \$8.25 for each reduced-price ticket; thus, the sales for reduced-price tickets, in dollars, is given by $8.25r$. Therefore, the total sales, in dollars, for the movie showing is given by $11f + 8.25r$. Since the total sales for all full-price and reduced-price tickets is \$2,145, it follows that $11f + 8.25r = 2,145$.

Choice A is incorrect. This system of equations suggests that the movie theater sold a total of 2,145 full-price and reduced-price tickets for a total of \$214. Choice C is incorrect. This system suggests that the movie theater charges \$8.25 for each full-price ticket and \$11 for each reduced-price ticket. Choice D is incorrect. This system suggests that the movie theater charges \$8.25 for each full-price ticket and \$11 for each reduced-price ticket and sold a total of 2,145 tickets for a total of \$214.

Q6**A**

A. $f + h \leq 14$

Choice A is correct. It's given that the cleaning service cleans both offices and homes, where f is the number of offices and h is the number of homes the cleaning service can clean per day.

Therefore, the expression $f + h$ represents the number of places the cleaning service can clean per day. It's also given that the cleaning service can clean at most 14 places per day. Since $f + h$ represents the number of places the cleaning service can clean per day and the service can clean at most 14 places per day, it follows that the inequality $f + h \leq 14$ represents this situation.

Choice B is incorrect. This inequality represents a cleaning service that cleans at least 14 places per day.

Choice C is incorrect. This inequality represents a cleaning service that cleans at most 14 more offices than homes per day.

Choice D is incorrect. This inequality represents a cleaning service that cleans at least 14 more offices than homes per day.

Q7**D**

D. $y = -5x + 18$

Choice D is correct. A linear relationship can be represented by an equation of the form $y = mx + b$, where m and b are constants. It's given in the table that when $x = 0$, $y = 18$. Substituting 0 for x and 18 for y in $y = mx + b$ yields $18 = m0 + b$, or $18 = b$. Substituting 18 for b in the equation $y = mx + b$ yields $y = mx + 18$. It's also given in the table that when $x = 1$, $y = 13$. Substituting 1 for x and 13 for y in the equation $y = mx + 18$ yields $13 = m1 + 18$, or $13 = m + 18$. Subtracting 18 from both sides of this equation yields $-5 = m$. Therefore, the equation $y = -5x + 18$ represents the relationship between x and y .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Q8**8.6**

Accept also: 43/5

The correct answer is 8.6. Dividing both sides of the given equation by 10 yields $x = 8.6$. Therefore, 8.6 is the solution to the given equation. Note that 8.6 and $43/5$ are examples of ways to enter a correct answer.

Q9**A**

A. -34

Choice A is correct. It's given that the function g is defined by $gx = 4x - 6$. Substituting -7 for x into the given equation yields $g-7 = 4(-7) - 6$, or $g-7 = -34$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the value of x for which $gx = -7$, not the value of $g-7$.

Q10**30**

The correct answer is 30. It's given that $x + y = 125$. Substituting 125 for $x + y$ in the second equation of the given system yields $125 + y = 155$. Subtracting 125 from both sides of this equation yields $y = 30$. Therefore, the value of y is 30.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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